ClusterFusion

Project

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Overview

Text Summarization is the process of condensing a large text into a shorter version while retaining its key information. It helps extract the most important points from lengthy documents, making them easier to read and understand . The process involves analyzing the text, identifying important sentences or keywords, and generating a concise summary.

Steps of My Project/Key Features

Text Preprocessing:

- Breaking the document into sentences or words (**Tokenization**).
- Removing unnecessary words (**Stopwords**) and normalizing the text.

- **Tokenization**: The text is divided into sentences.
- **Stop Word Removal**: Common, less meaningful words like "the," "and," or "is" are removed.
- **Word Frequency Analysis**: The tool calculates the importance of words by counting their frequency.
- **Sentence Scoring**: Each sentence is scored based on its meaningful words' frequency.
- **Summary Generation**: Sentences with scores above a calculated threshold are selected for the summary.

Key Features:

- Uses **TF-IDF** logic to identify important words.
- Avoids unimportant or repetitive sentences.
- Implements algorithms for punctuation removal, word conversion to lowercase, and sentence scoring.

```
double tf(const vector<string>& doc, const string& term) {
        double count = 0;
        for (const auto& word : doc) {
                if (term == word) {
                        count++;
                }
        }
        return count / doc.size();
```

```
double idf(const vector<vector<string>>& docs, const string& term) {
        double n = 0;
        for (const auto& doc : docs) {
                if (find(doc.begin(), doc.end(), term) != doc.end()) {
                        n++;
                }
        }
        if (n == 0) {
                return 0;
        }
        return log(docs.size() / n);
}
```

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```
void scoreSentences() {
        for (const string& sentence :
sentences) {
            double sentenceScore = 0;
            int noOfTotalWords =
spaceCounter(sentence) + 1;
            int noOfNonStopWords =
countNonStopWords(sentence);

            sentenceScore =
static_cast<double>(noOfNonStopWords) /
noOfTotalWords;
            sentencesScore[sentence] =
sentenceScore;
        }
    }
```

Summary:

Freelance writer Sng Ler Jun has been taking on more writing assignments since March to earn extra money for his family and himself.

Battling pandemic fatigue: Some feel burned out as the Covid-19 outbreak drags on with no end in sight.

Application

1. **News Aggregation**
2. **Search Engines**
3. **Educational Tools**
4. **Customer Support**
5. **Social Media**
6. **Healthcare**
7. **E-Learning and AI Assistant**

Next Key Features :

- Text Rank Algorithms
- Direct Method Algorithm

http://github.com/morshaline1504